

Ricardo (Zhicong) Xie

Simulation / Analysis Intern | FEA, CFD and Validation

Auckland, New Zealand | +64 29 027 29390 | zxie588@aucklanduni.ac.nz

LinkedIn: <https://www.linkedin.com/in/ricardo-xie-6394643b4> | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

NZ student visa | Full-time eligible over scheduled summer break 17 Nov 2026 - late Feb 2027 | No employer visa support needed for this internship | Open NZ-wide

PROFILE

- Mechanical master's student focused on structural FEA, thermal-fluid CFD and model credibility checks.
- Deliverables include load cases, boundary conditions, mesh checks, balance review and model limits.
- Project screenshots and source links: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Study focus and portfolio work in mechanical, building-services, manufacturing and analysis contexts.

Study record/focus: Completed: Computational Fluid Dynamics, Composite Materials, Engineering Methods, Aerospace Systems Engineering.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Engineering team leadership: led task scoping, workload split and final technical presentation in team design work.
- Workshop practice: prepared AutoCAD/DXF laser-cut geometry and checked cut parts against drawings.
- Hands-on exposure: CNC machining, workholding, cutting fluid, chip formation and 3D printing workflow.
- Lab practice: interpreted force-displacement tests and controlled a robotic arm in training sessions.
- Technical communication: former student news lead; strong project records, photography and presentation visuals.

PROJECT WORK

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Built an Abaqus shell FEA model with S4R elements, laminate layups and local mesh refinement.
- Converted aero, lateral cornering and 4.1 G bump conditions into structural load cases.
- Checked ply-level stress use; peak utilisation was about 74% under the 4.1 G bump case.

Thermal-Fluid Simulation & Verification - Solar Chimney Case Study

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Assessed model credibility through mesh sensitivity, mass/heat balance and Reynolds/Rayleigh scaling.
- Documented model boundaries, limitations and recommended physical data points needed for future experimental correlation.

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, testing workflow

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Reviewed load paths around corner, strap-lug and screw-boss regions; built Keil C module structure.

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Built a small NZ residential MVHR/ERV coordination package with Revit MEP, SolidWorks and AutoCAD/DXF evidence.
- Created plan, 3D view, airflow schedule, sheet, review comments and simple clash/coordination notes.
- Documented preliminary airflow, pressure-loss, balancing and commissioning actions for the next design iteration.

TECHNICAL SKILLS

- FEA/CFD: Abaqus/Standard, ANSYS Fluent/CFX, shell FEA and conjugate heat transfer.
- Model setup: load cases, boundary conditions, mesh checks, assumptions log and result limits.
- Validation: composite ply utilisation, Rayleigh/Reynolds scaling and mass/heat balance checks.